

LABORATORY GUIDE

AUTONOMOUS SYSTEMS

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Custódio

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PRE-REQUISITES: Laboratory computers

- Ubuntu with ROS 2 Humble Hawksbill pre-installed.
- Computers are connected to the same network (deec-robots).
- LSDC4 (laboratory) desktop account credentials:
 - Username: saut
 - Password: saut_alunos

INSTALLING LINUX AND ROS 2

- Each ROS 2 version is linked to an Ubuntu LTS release:

Operating System	ROS 2 Versions	Python Version
<u>Ubuntu 18.04 LTS</u> (not recommended)	Eloquent Elusor (among others)	3.6
<u>Ubuntu 20.04 LTS</u> (not recommended)	Foxy Fitzroy (and Galactic Geochelone)	3.8
<u>Ubuntu 22.04 LTS</u>	<u>Humble Hawksbill</u>	3.10
<u>Ubuntu 24.04 LTS</u>	<u>Jazzy Jalisco</u>	3.12

- Despite the selected version, ROS 2 desktop-full install is recommended.

RECOMMENDATIONS

1. USE TERMINATOR

- On the laptop, run

```
sudo apt-get install terminator
```

Tips and Tricks

```
ctrl + shift + C (copy)
ctrl + shift + V (paste or use the right button of mouse)
ctrl + shift + O (split the terminal horizontally)
ctrl + shift + E (split the terminal vertically)
ctrl + shift + Z (focus/unfocus a terminal window)
```



2. DEVELOP YOUR PROJECT IN PYTHON 3

GETTING STARTED WITH ROS2



[Learn ROS 2: Beginner to Advanced Course \(Concepts and Code\), by Kevin Wood](#)

THE LABORATORY

- Located at the 5th floor of the North Tower – Room LSDC4



WIFI (Pioneers, Alfabots and Turtlebots):

Network: deec-robots

Password: shakeytherobot

IP range: 10.16.140.<id>

Lab computers:

Username: saut

Password: saut_alunos

THE LAB COMPUTERS

- Lab computers shall be used to test code or to record bag files and **NOT for development**. **Do not** delete files inside robots, except your own files (i.e. bags).
- Check the Wi-Fi connection of the lab computers to deec-robots network (if not working properly, please contact Mr. Manuel Ribeiro (office 5.18, mrribeiro@tecnico.ulisboa.pt, phone no. 2189 inside Técnico, 218418189 from outside).
- Please, **do NOT** change the ~/.bashrc file configuration of the robots (only in PCs).
- To work with a specific robot, you must export its respective **ROS_DOMAIN_ID**. The ROS_DOMAIN_ID corresponds to each robot's <id> (according to **10.16.140.<id>**):

```
export ROS_DOMAIN_ID=<id> # execute this command on every new terminal session
```

THE PIONEER ROBOTS

- The Pioneer robots are configured with:
 1. A Raspberry Pi with Ubuntu 22.04LTS and ROS2 Humble
 2. The P2OS Package, [refactored for ROS2](#), originally provided for [ROS1](#)
- Pioneer node **/p2os_driver**
 - Subscribes:
 - **/cmd_vel** (geometry_msgs/msg/Twist), etc.
 - Publishes:
 - **/odom** (nav_msgs/msg/Odometry),
 - **/sonar_[0...15]/range** (sensor_msgs/msg/Range), etc.
- Hokuyo laser rangefinder node **/urg_node2**
 - Publishes:
 - **/scan** (sensor_msgs/msg/LaserScan)



CREDENTIALS:

Username: deec

Password: deecrobots

Network: deec-robots

Password: shakekeytherobot

IP range: 10.16.140.[17...23]

THE PIONEER ROBOTS [Base Workspace]

1. The [Pioneer packages](#) are open-source and provided as a base workspace; in your laptop, do

```
mkdir -p ~/pioneer_ws/src  
cd ~/pioneer_ws/src
```

2. Clone the repository

```
git clone https://github.com/JPLDevMaster/p2os2.git  
cd p2os2  
git checkout p3dx-enabled
```

3. Build the workspace with [colcon](#)

```
cd ~/pioneer_ws  
colcon build --symlink-install
```

4. Source the workspace in new terminals

```
source install/setup.bash
```

THE PIONEER ROBOTS [Connecting]

1. On a new terminal of the Laptop/Lab computer, ping the RPi of the Pioneer once turned on

```
ping 10.16.140. [17...23]
```

2. After a successful ping attempt, ssh into the machine

```
ssh deec@10.16.140. [17...23]
```

3. Launch the driver

```
ros2 launch p2os_bringup p2os_driver_launch.py
```

4. On a new terminal, SSH into the machine again (step 2), and verify environment

```
ros2 topic list
```

5. Verify the motors are enabled; if the output is 0, press the MOTORS button on the robot

```
ros2 topic echo /motor_state
```

If ipconfig is not available, use ip address or install net-tools to run ifconfig:
apt update
apt install net-tools

THE PIONEER ROBOTS [Communications]

1. Run in the Laptop/Lab computer (or append to your ~/.bashrc file) [*or SSH into machine*] [*]

```
export ROS_DOMAIN_ID=[17...23]
```

2. Check available topics

```
ros2 topic list
```

- 3a. Publish motion commands; move-forward example:

```
ros2 topic pub --rate 10 /cmd_vel geometry_msgs/msg/Twist "{linear: {x:  
0.1}, angular: {z: 0.0}}"
```

- 3b. Moving alternative is keyboard; in a new terminal, execute the [teleop](#) node

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

[*] certain local network configurations might forbid the export pipeline; in such cases, use the SSH alternative

THE PIONEER ROBOTS [Communications]

4. Once familiarized, [record](#) data

```
ros2 bag record -a (for all topics)
ros2 bag record <topic_name> (recommended: for specific topics)
```

5. If the bag was recorded in the machine (in <dir_name>), copy it to your local machine

```
scp -r deec@10.16.140.[17...23]:/<dir_name> .
```

6. *[in SSH]* If the bag was recorded in the machine, delete the recording and return

```
rm -rf ~/<dir_name>
exit
```

THE PIONEER ROBOTS [Sonar & Hokuyo rangefinder]

1. Connect the rangefinder to the machine via USB. Then, SSH into the robot

```
ssh deec@10.16.140. [17...23]
```

2. Launch the rangefinder driver

```
ros2 launch urg_node2 urg_node2.launch.py
```

3. On a new terminal, verify new node and topic

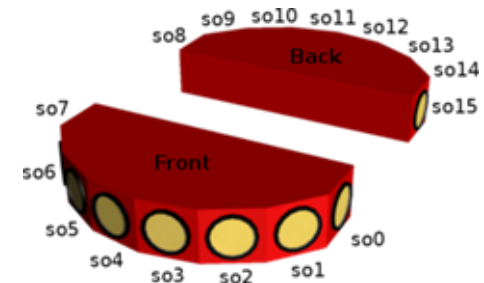
```
ros2 topic echo /scan
```



Hokuyo: request access to
Mr. Manuel Ribeiro

1. The Pioneer robots dispose of built-in sonar sensors [*]:

```
ros2 topic echo /sonar_[0...15]/range [16 different ranges]
```



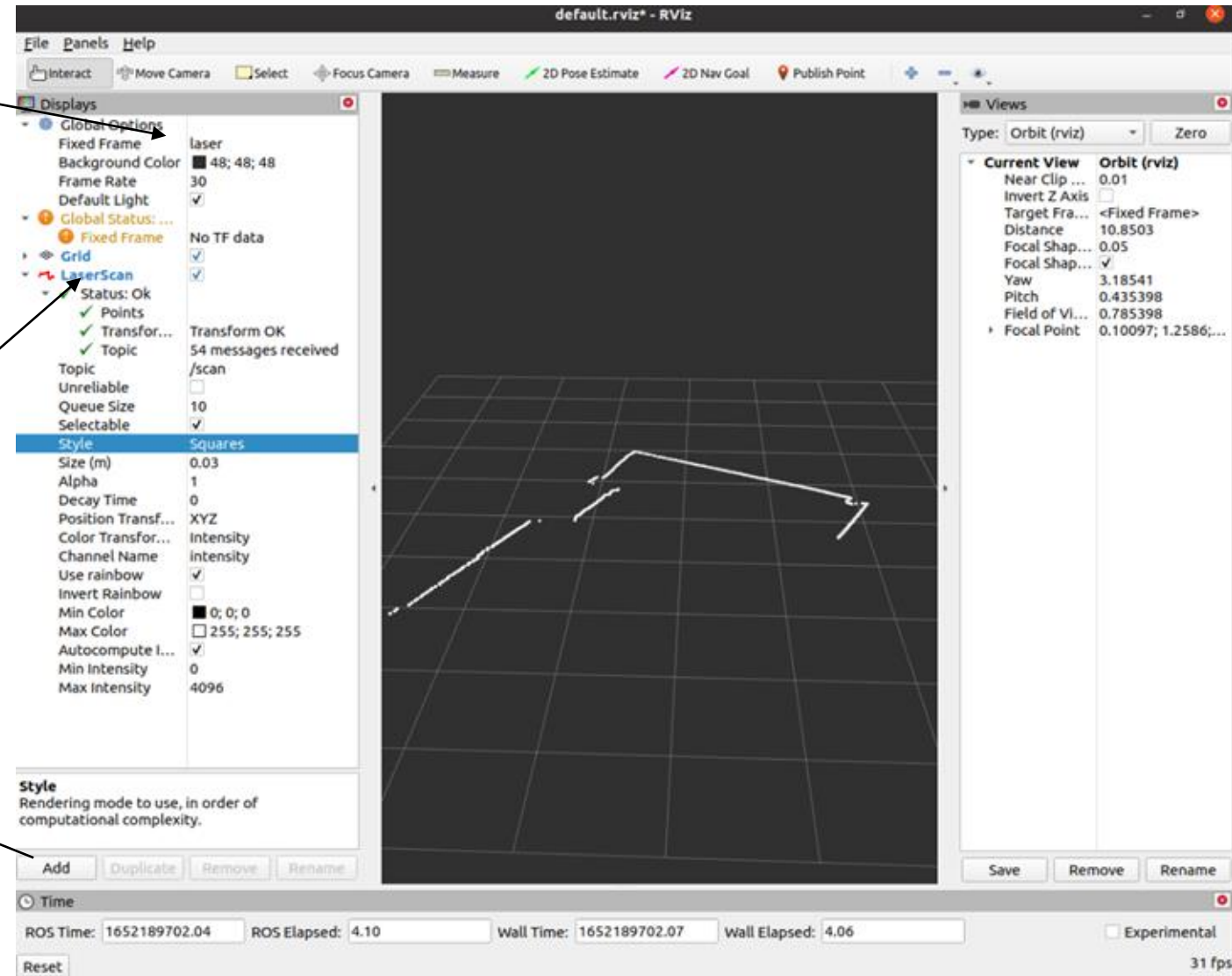
[*] not all pioneers dispose of fully-operational sonar capabilities; not all pioneers have front **and** back sensors

[source](#)

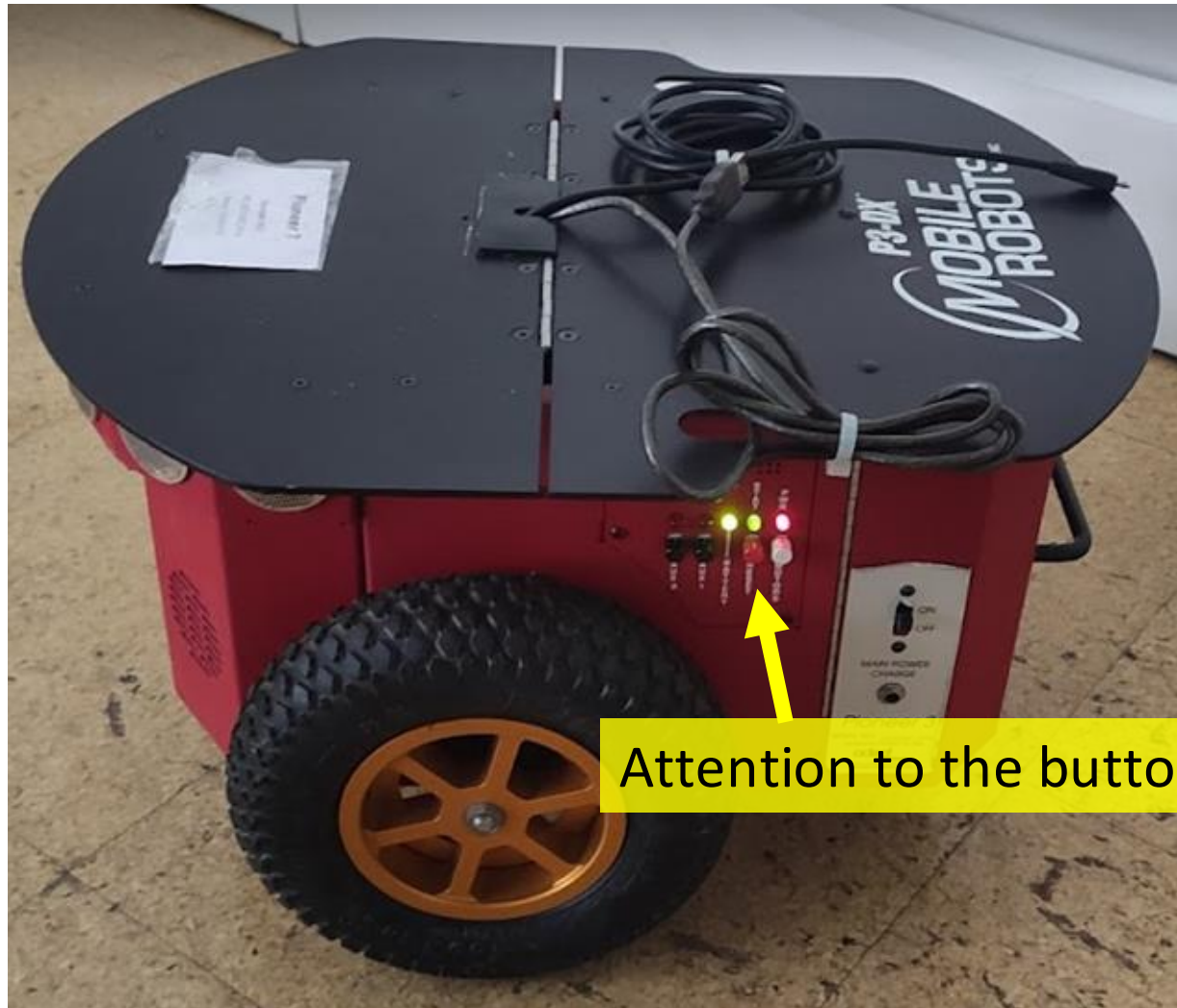
THE PIONEER ROBOTS [Hokuyo rangefinder]

Write “laser”, since no referential was defined.

Add “LaserScan” node



THE PIONEER ROBOTS [Charging]



Attention to the buttons!



THE TURTLEBOT3 ROBOTS

- The Turtlebots 3 are configured with a Raspberry Pi with Ubuntu 22.04LTS and ROS 2 Humble
- Relevant topics:
 - Subscribes:
 - `/cmd_vel` (geometry_msgs/msg/Twist)
 - Publishes:
 - `/odom` (nav_msgs/msg/Odometry)
 - `/tf` (tf2_msgs/msg/TFMessage)
 - `/scan` (sensor_msgs/msg/LaserScan)
- Additional resources:
 - <https://emanual.robotis.com/docs/en/platform/turtlebot3/quick-start/>



CREDENTIALS:

Username: deec

Password: deecrobots

Network: deec-robots

Password: shakeytherobot

IP range: 10.16.140.[11...15]

THE TURTLEBOT3 ROBOTS [Connecting]

1. On a new terminal of the Laptop/Lab computer, and SSH into the machine

```
ssh deec@10.16.140. [11...15]
```

2. Launch the robot drivers

```
ros2 launch turtlebot3_bringup robot.launch.py
```

THE TURTLEBOT3 ROBOTS [Laptop Setup]

3. Install the Turtlebot3 packages - run in the **laptop** once [**skip for lab computers**].

```
sudo apt-get update
sudo apt-get install ros-humble-dynamixel-sdk
sudo apt-get install ros-humble-turtlebot3-msgs
sudo apt-get install ros-humble-turtlebot3
sudo apt-get install ros-humble-teleop-twist-keyboard
```

If `ipconfig` is not available, use `ip` address or install `net-tools` to run `ifconfig`:
`apt update`
`apt install net-tools`

4. Additional configurations. Run in the **Laptop/Lab computer** (or append to you `~/.bashrc` file).

```
export TURTLEBOT3_MODEL=waffle_pi
export ROS_DOMAIN_ID=<[11...15]>
```

[TurtleBot3 DOMAIN ID]

THE TURTLEBOT3 ROBOTS [Communications]

5. On a new terminal of the Laptop, test the communications: can the PC show the topics?

```
ros2 topic list
```

6. Interesting topics to query: /odom or /scan

```
ros2 topic echo /odom  
ros2 topic info /odom  
ros2 topic hz /odom
```

7. Test motion commands

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0},  
angular: {x: 0.0, y: 0.0, z: 0.5}}" (or)  
ros2 topic pub --rate 1 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.1, y: 0.0,  
z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

8. To record topics in rosbags *

```
ros2 bag record -a (for all topics)  
ros2 bag record <topic name> (for specific topics)
```

* If path unspecified, bags get recorded in the folder where the command is run. For more details, click [here](#).

THE TURTLEBOT3 ROBOTS [Communications]

THIS APPLIES ONLY when the laptop/lab computer does not see the robot's topics

5. [ONLY IF STEP 5 DID NOT WORK] SSH into the machine and check available topics again

```
ssh deec@10.160.140. [11...15]  
ros2 topic list
```

6. Interesting topics to query: /odom or /scan

```
ros2 topic echo /odom  
ros2 topic info /odom  
ros2 topic hz /odom
```

7. Test motion commands

```
ros2 topic pub --rate 1 /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.1, y: 0.0,  
z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

8. To record topics in rosbags

```
ros2 bag record -a (for all topics)  
ros2 bag record <topic name> (for specific topics)
```

9. Using the bag recorded in the machine. copy it (in <dir_name>) to your local machine

```
scp -r deec@10.16.140. [11...15]:/<dir_name> .
```

10. [in SSH] Delete the bag recording and return

```
rm -rf ~/<dir_name>  
exit
```

THE TURTLEBOT3 ROBOTS [Operation]

11. Control the robot. On the laptop/Lab computer, run

```
ros2 run turtlebot3_teleop teleop_keyboard
```

Accessing the robot camera

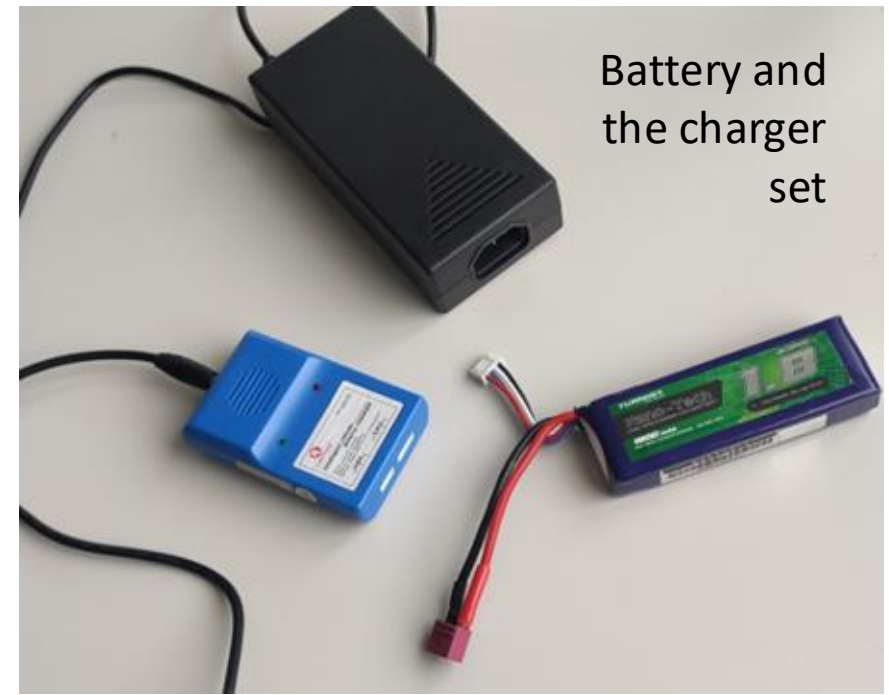
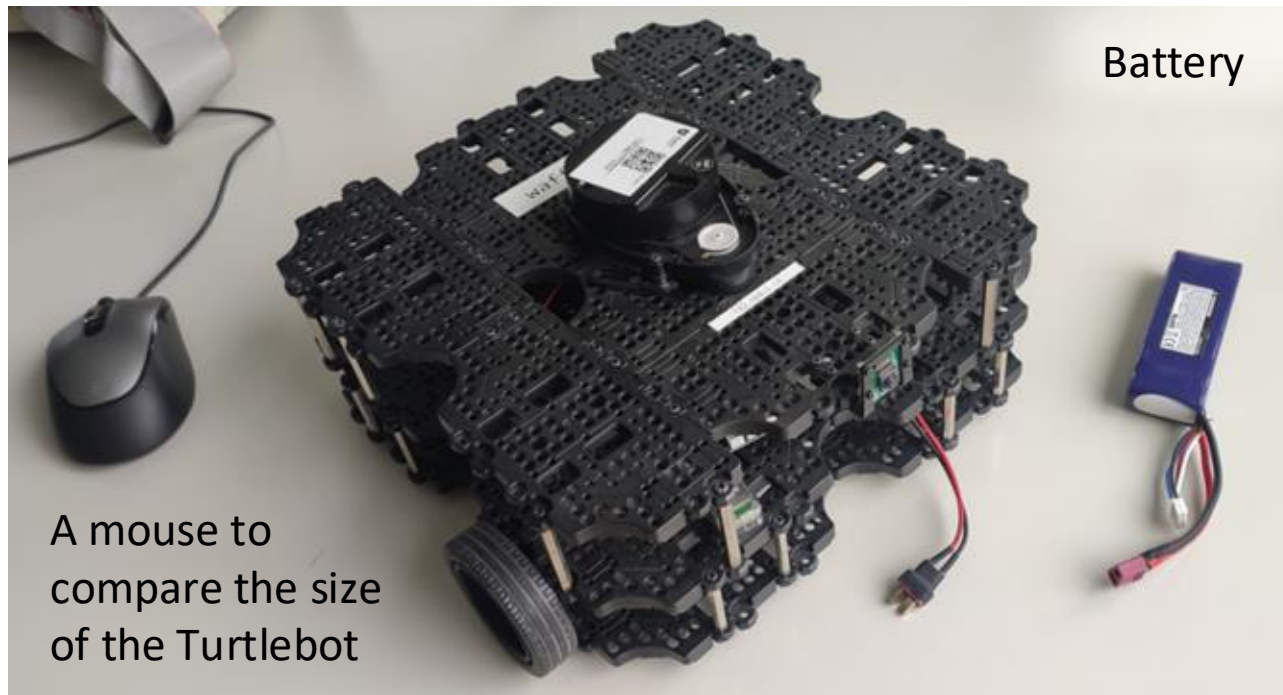
1. Launch the camera node on the robot

```
ssh deec@10.16.140. [11...15]  
ros2 run v4l2_camera v4l2_camera_node --ros-args -p image_size:="[320,240]"
```

2. To see the camera feed, run on the laptop/Lab computer

```
ros2 run rqt_image_view rqt_image_view
```

THE TURTLEBOT ROBOTS [Charging]



THE TURTLEBOT4 ROBOTS

- The Turtlebots 4 are configured with a Raspberry Pi 4B with Ubuntu 24.04 LTS and ROS 2 Jazzy.
- Relevant topics:
 - Subscribes:
 - `/cmd_vel_unstamped`
(geometry_msgs/msg/Twist)
 - Publishes:
 - `/odom` (nav_msgs/msg/Odometry)
 - `/tf` (tf2_msgs/msg/TFMessage)
 - `/scan` (sensor_msgs/msg/LaserScan)
- Additional resources:
 - <https://turtlebot.github.io/turtlebot4-user-manual/>

Warning

ROS 2 Jazzy is not installed on the Lab PCs, only ROS 2 Humble



CREDENTIALS:

Username: deec

Password: deecrobots

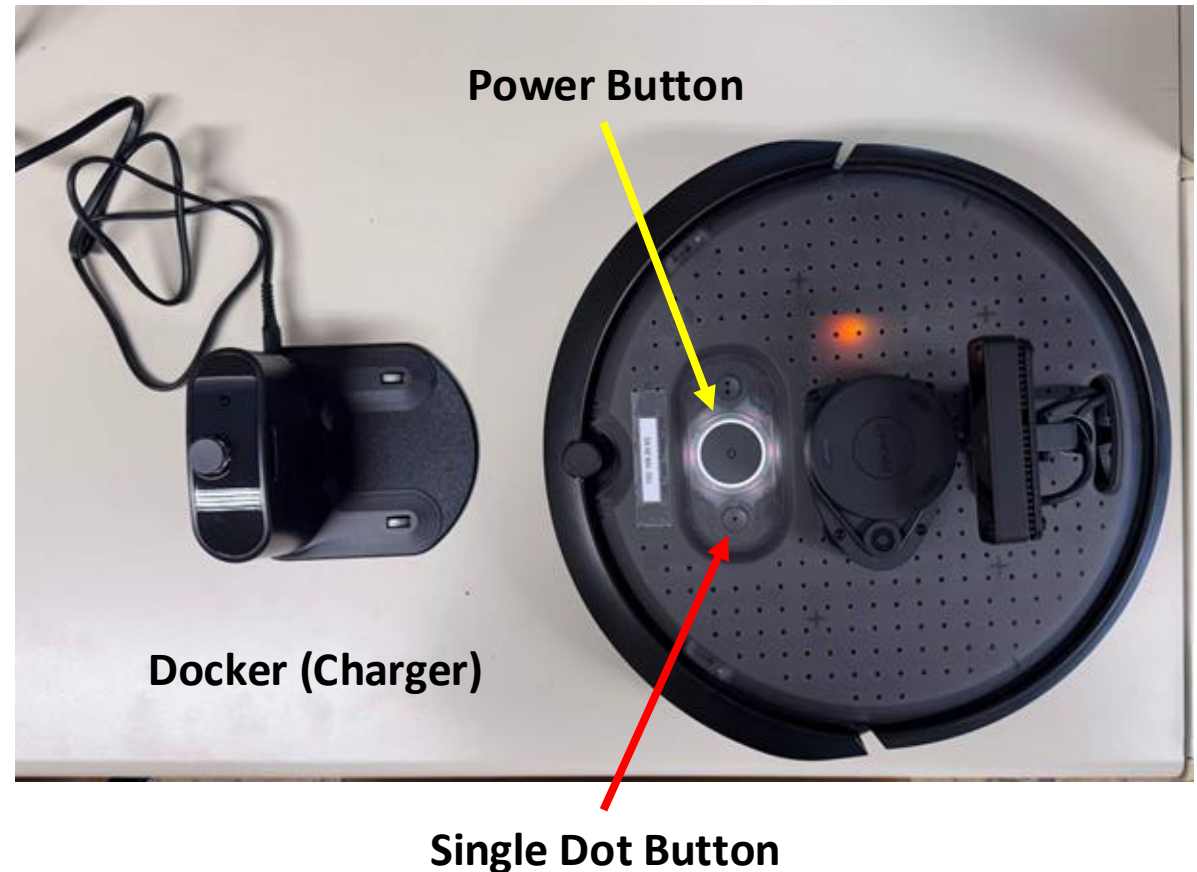
Network: deec-robots

Password: shakeytherobot

IP range: 10.16.140.[81...85]

THE TURTLEBOT4 ROBOTS [Turn On/Charging]

- Place and connect the robot to a docker to turn it on.
- Hold the **power button** (while off the docker) for 7 seconds to turn off the robot.
- Press the **single dot button** for 10 seconds to go into standby mode (automatic when placed on the docker, for a long period - “charging mode”).



For more details check:

https://iroboteducation.github.io/create3_docs/hw/face/

THE TURTLEBOT4 ROBOTS [Laptop Setup]

1. Install the TurtleBot4 packages - run in your laptop once.

```
sudo apt-get update && sudo apt install ros-jazzy-turtlebot4-desktop  
sudo apt-get install ros-jazzy-turtlebot4-description  
sudo apt-get install ros-jazzy-turtlebot4-msgs  
sudo apt-get install ros-jazzy-turtlebot4-navigation  
sudo apt-get install ros-jazzy-turtlebot4-node  
sudo apt-get install ros-jazzy-teleop-twist-keyboard
```

2. Upgrade your system [Optional].

```
sudo apt-get upgrade
```

3. Additional configurations. Run in your Laptop (or append to your ~/.bashrc file).

```
export ROS_DOMAIN_ID=<[81...85]>
```

[TurtleBot4 DOMAIN ID]

THE TURTLEBOT4 ROBOTS [Communications]

4. Test the communications: can the computer see the robot's topics?

```
ros2 topic list
```

5. Interesting topics to query: /odom or /scan

```
ros2 topic echo /odom  
ros2 topic info /odom  
ros2 topic hz /odom
```

6. Test motion commands

```
ros2 topic pub --rate 1 /cmd_vel_unstamped geometry_msgs/msg/Twist "{linear: {x:  
1.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 5.0}}"
```

7. To record topics in rosbags *

```
ros2 bag record -a (for all topics)  
ros2 bag record <topic-names> (for specific topics)
```

* If path unspecified, bags get recorded in the folder where the command is run. For more details, click [here](#).

THE TURTLEBOT4 ROBOTS [Operation]

8. Control the robot. On your laptop (with ROS 2 Jazzy installed), run

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args -p stamped:=false  
--remap cmd_vel:=/cmd_vel_unstamped
```

Accessing the robot camera

To check the camera feed, run

```
ros2 topic echo /oakd/rgb/preview/image_raw  
ros2 topic info /oakd/rgb/preview/image_raw  
ros2 topic hz /oakd/rgb/preview/image_raw  
ros2 run rqt_image_view rqt_image_view
```

THE ALPHABOT2 ROBOTS

- The Alphas are configured with a Raspberry Pi 3 B, Ubuntu 22.04 LTS and ROS 2 Humble Hawksbill.
- Relevant topics:
 - Subscribes:
 - `/cmd_vel` (geometry_msgs/msg/Twist)
 - Publishes:
 - `/camera/compressed` (sensor_msgs/msg/CompressedImage)
 - `/virtual_odometry` (nav_msgs/msg/Odometry)
- Additional resources:
 - [ROS & Python drivers](#)

Warning

These robots do **NOT** have (actual) wheel odometry, IMU or LiDAR



CREDENTIALS:

Username: deec

Password: deecrobots

Network: deec-robots

Password: shakekeytherobot

IP range: 192.168.28.[50..70]

THE ALPHABOT2 ROBOTS

1. Open a terminal on a Laptop/Lab computer, and SSH into the RPi of the **Alphabot**.

```
ssh deec@10.16.140. [50...70]
```

2. Launch AlphaBot2 drivers inside the robot:

```
ros2 launch alphabot2 alphabot2_launch.py
```

3. In a new terminal, initialize motion drivers:

```
ssh deec@10.16.140. [50...70]  
ros2 run alphabot2 motion_driver
```

4. In a new terminal, run in the **Laptop/Lab computer** (or append to you ~/.bashrc file):

```
export ROS_DOMAIN_ID=< [50...70] > [AlphaBot DOMAIN ID]
```

THE ALPHABOT ROBOTS [Communications]

5. In the same terminal, test the communications: can the PC showcase the robot's topics?

```
ros2 topic list
```

6. In the same terminal, try to move the robot*:

```
ros2 topic pub --rate 1 /alphabot2/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 1.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 5.0}}"
```

[pub topic]
[or teleop *]

```
ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

7. Interesting topics to query: `/image/compressed`

```
ros2 topic echo /image/compressed
```

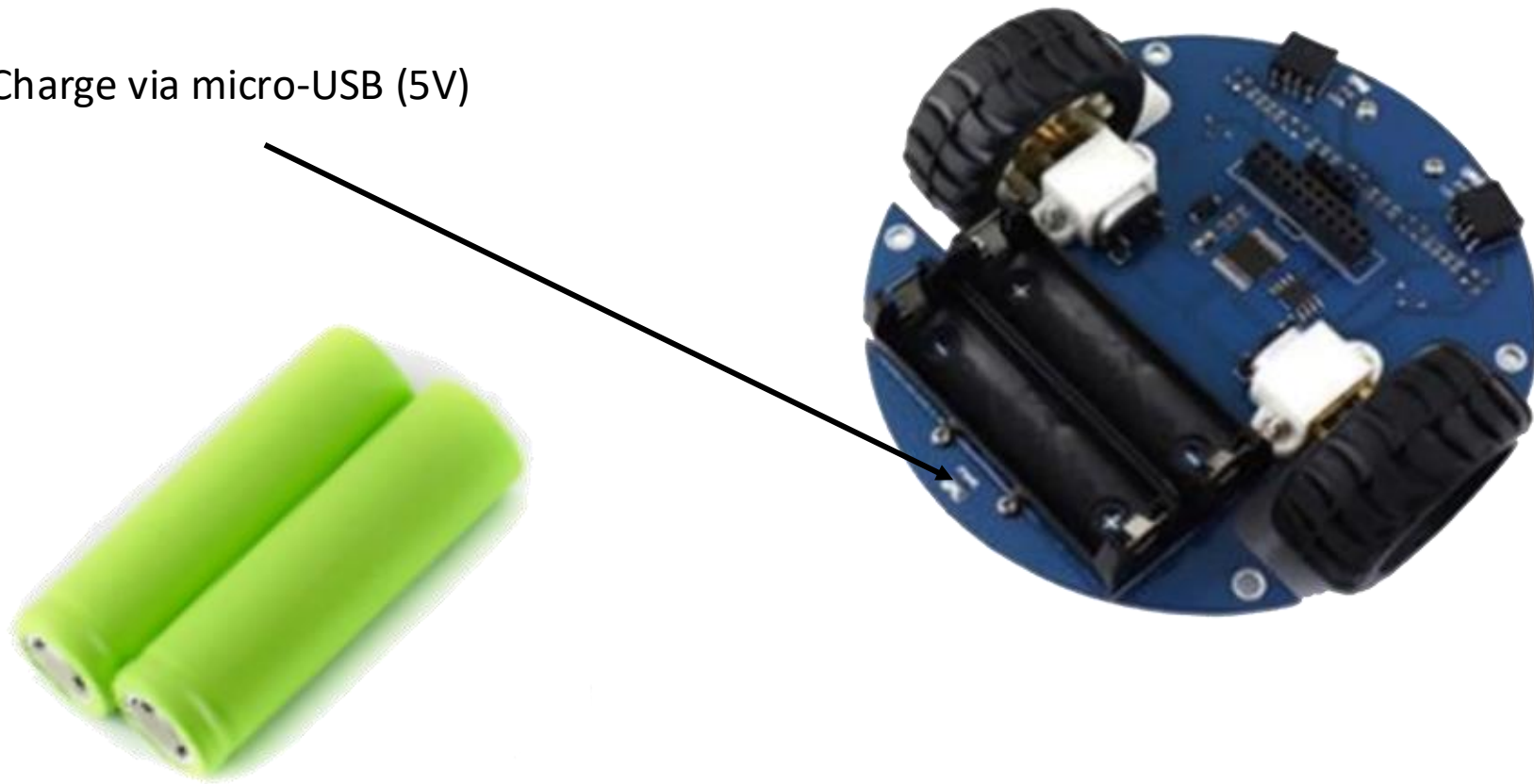
8. To see the camera feed, run on the laptop/Lab computer

```
ros2 run rqt_image_view rqt_image_view
```

*You can also run this commands on your Laptop/Lab PC, after completing step 5., i.e. exporting `ROS_DOMAIN_ID`. If the teleop package is not installed, run: `sudo apt install ros-humble-teleop-twist-keyboard`

THE ALPHABOT ROBOTS [Charging]

Charge via micro-USB (5V)



ROS CHEAT-SHEET

Relevant terms to search

Package	Bag
Node	Launch Files
Topic	Parameters
Publisher	TF
Subscriber	RVIZ
Message	Gazebo

Graphical user tools

```
rviz2
```

```
rqt (can be used to monitor)
```

```
ros2 run rqt_tf_tree rqt_tf_tree
```

```
ros2 run rqt_plot rqt_plot
```

Terminal user tools

```
ros2 launch <package_name> <launch_file>  
ros2 run <package_name> <executable>  
ros2cd <package_name>
```

```
ros2 topic list  
ros2 topic info <topic_name>  
ros2 topic hz <topic_name>
```

```
ros2 node list  
ros2 node info <node_name>
```

```
ros2 bag record <topic-names>  
ros2 bag info <bag-name>  
ros2 bag play <bag-name>
```

```
ros2 service list  
ros2 service call <options>
```

```
ros2 msg list
```

ROS EXTRA NOTES

- Often forgotten – set the environment variables used by ROS

```
source devel/setup.bash # in the colcon workspace dir
```

- Confirm the definition of the ROS packages path

```
echo $ROS_PACKAGE_PATH
```

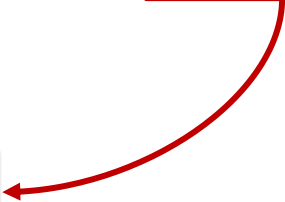
- ROS/Linux editors: vim or nano [nano is recommended]
- Make use of the .bashrc file

```
source /opt/ros/noetic/setup.bash  
source ~/catkin_ws/devel/setup.bash  
export EDITOR='nano -w'
```

- Copy a file from the robot to your laptop/Lab computer

```
scp <robot_username>@<robot_ip>:<path_to_file> <location_in_your_computer>
```

IMPORTANT



- Explore the use of `ros2 launch` and launch files
- Bag files may occupy too much space.
Suggestion: record only the required topics and include compression
- If problems are detected, run

```
ros2 doctor -r
```
- In rviz2 you might need to adjust the option **Topic > Reliability Policy** to **Best Effort**

Extra software suggestions (cont.)

- **Foxglove** is a Rviz alternative, with data management and timeline visualization capabilities.

Available for Linux,
Windows and macOS.

<https://foxglove.dev/>



- Select a **physical robot** for a first trial
- Connect to the robot and **list topics**
- **Visualize in real time sensor data** (e.g. Rviz, or Foxglove)
- **Move the robot** using the keyboard, while visualizing sensor data
- **Record a bag**, transfer it to an external computer, and visualize the data with **rosbag play**